

ECE321

Electronic circuits

Lecture (01)- Course Information

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- ☐ **Course Information**
- ☐ **Course Assessment**
- ☐ **Course Content**
- ☐ **Course Long-Term-Vision**
- ☐ **Time_plan and Project**
- ☐ **References**

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Course Information



☐ Lecture's Time:

☐ Sunday @8.30:11.40

☐ Lecture's Place:

☐ 27520-A4

☐ Lectures Materials, Quiz, Assignment, and Project:

☐ Please, check LMS

☐ Contacts:

☐ Email: Ahmed.reda@alumni.sjtu.edu.cn



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☐ Final Exam [67%]

- ☐ You must get > **30%** of the paper grade or you will **fail** in the course regardless of your course work.
- ☐ The final exam will be MCQ/ Written

☐ Midterm Exam [13%]

- ☐ The midterm exam will be MCQ/Written

☐ Quiz (**6X**), Assignment (**1X**), Project (**1X**) [20%]

- ☐ Upload Assignment Report
- ☐ Upload Project Report



 **Forbidden**

Copy or cheat during **Exams**
and even doing the **Quiz** and
Assignment

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Course Content (1/2)



☐ Course Objective

- ☐ Overview of important related topics on *Electronic Circuits*. The course consists of the **Eight** modules.

☐ Module 1: Frequency Response

- ☐ DC operating point, DC/AC analysis of amplifier , Miller effect, Low/High frequency response, Total frequency response, Phase response, Device application..etc.

☐ Module 2: Operational Amplifier

- ☐ Modeling of Op-amp, Linear and non-linear applications, Imperfections, Circuit implementation....etc.

☐ Module 3: Feedback Circuit

- ☐ Types of feedback, advantages and disadvantages, Analysis of feedback circuit....etc

☐ Module 4: Signal Generator and Wave Shaping.

- ☐ Analog/Digital multi-vibrator, Linear/Non-linear Oscillator, Schmitt trigger, bi-stable ..etc

Course Content (2/2)



☐ **Module 5: Quick Review**

- ☐ BJT and MOSFET devices (Construction, Operation, Small signal model, DC/AC analysis)

☐ **Module 6: Current Mirrors**

- ☐ Different types of current mirrors using BJT or MOSFET (simple, cascode, wilson, ...)

☐ **Module 7: Differential Amplifiers**

- ☐ DC operation, large signal operation, common-mode and differential mode response,...

☐ **Module 8: Operational Amplifiers Structure**

- ☐ Different types of op-amps (simple single-stage, cascode, telescopic cascode, folded cascode, and two stage op-amp)

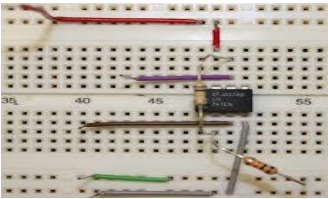
Instructor	Teaching Assistant
Dr. Ahmed Wahba & Dr. Ahmed Reda	Eng. Mahmoud & Eng. Asmaa



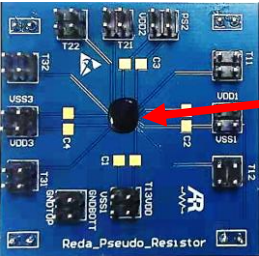
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Discrete Element



Bread Board



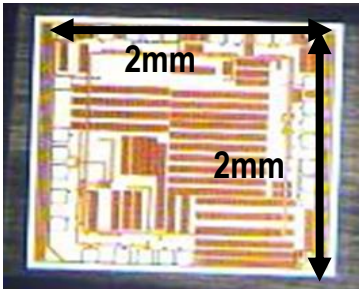
Chip-On-Board (COB)



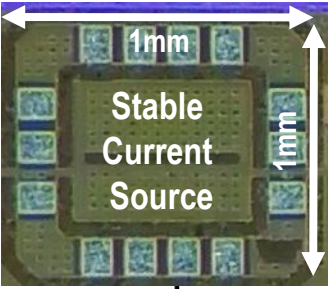
Printed Circuit Board (PCB)

- ☐ Through-hole
- ☐ Surface mount

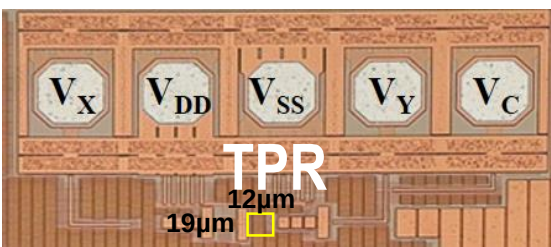
Vey Large Scale Integrated Circuit



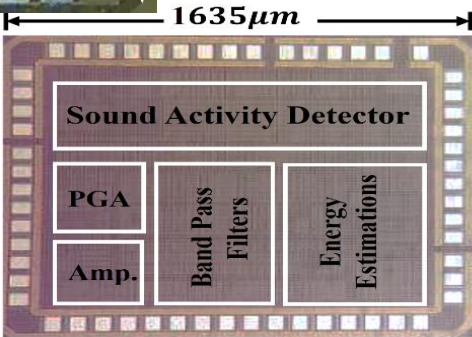
2mm
2mm



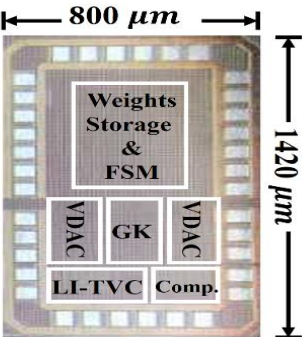
1mm
1mm
Stable Current Source



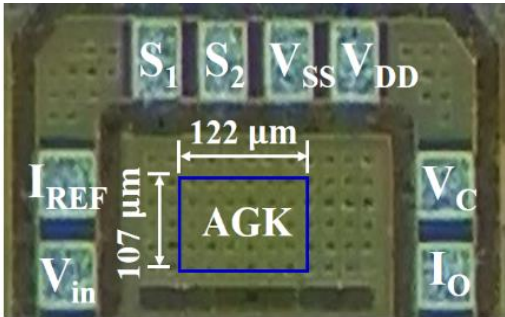
TPR
12µm
19µm



1635µm
800µm
Sound Activity Detector
PGA
Band Pass Filters
Energy Estimations
Amp.

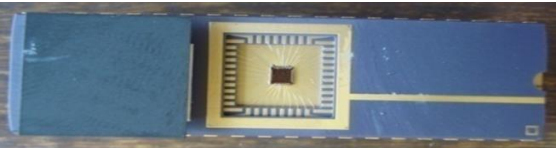


1420µm
800µm
Weights Storage & FSM
V_{DD}
GK
V_{DAC}
LI-TVC
Comp.



S₁ S₂ V_{SS} V_{DD}
122µm
107µm
AGK
I_{REF} V_{in} V_C I_O

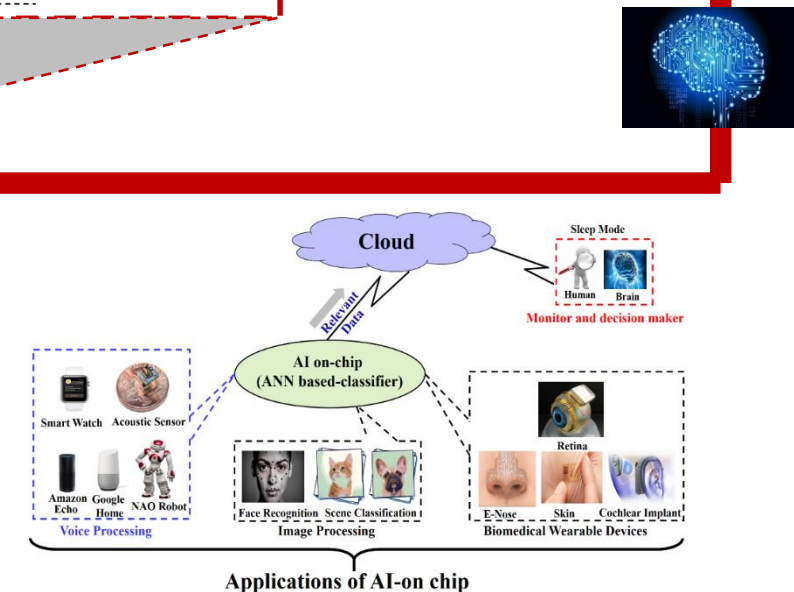
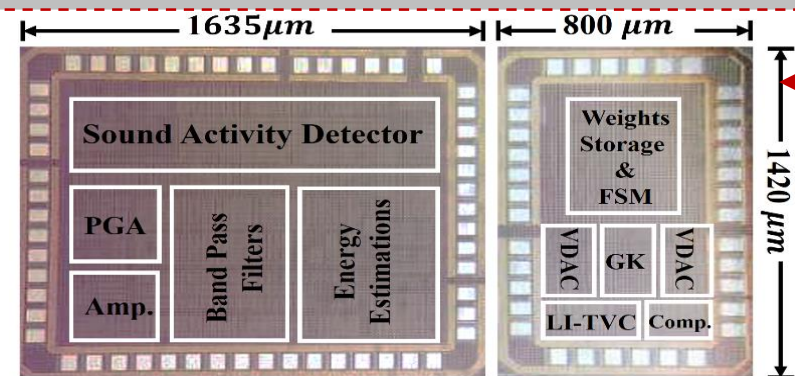
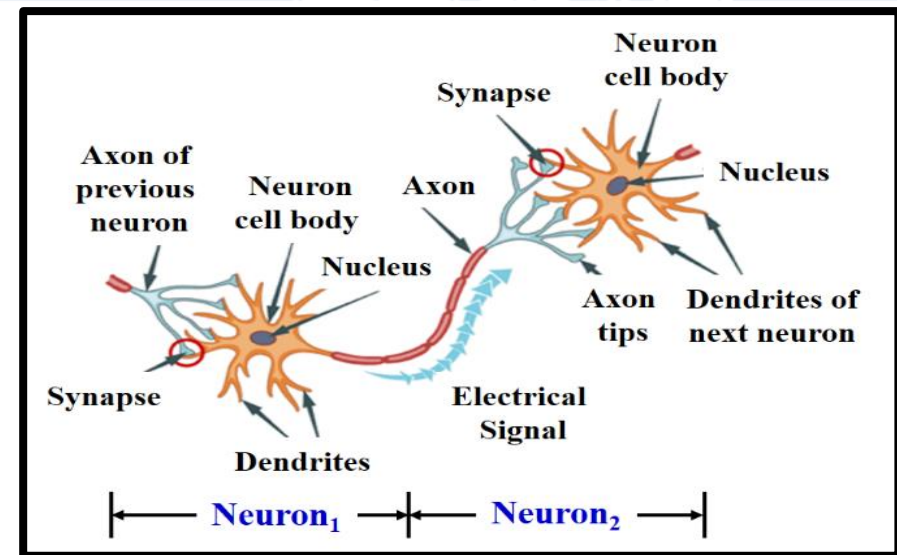
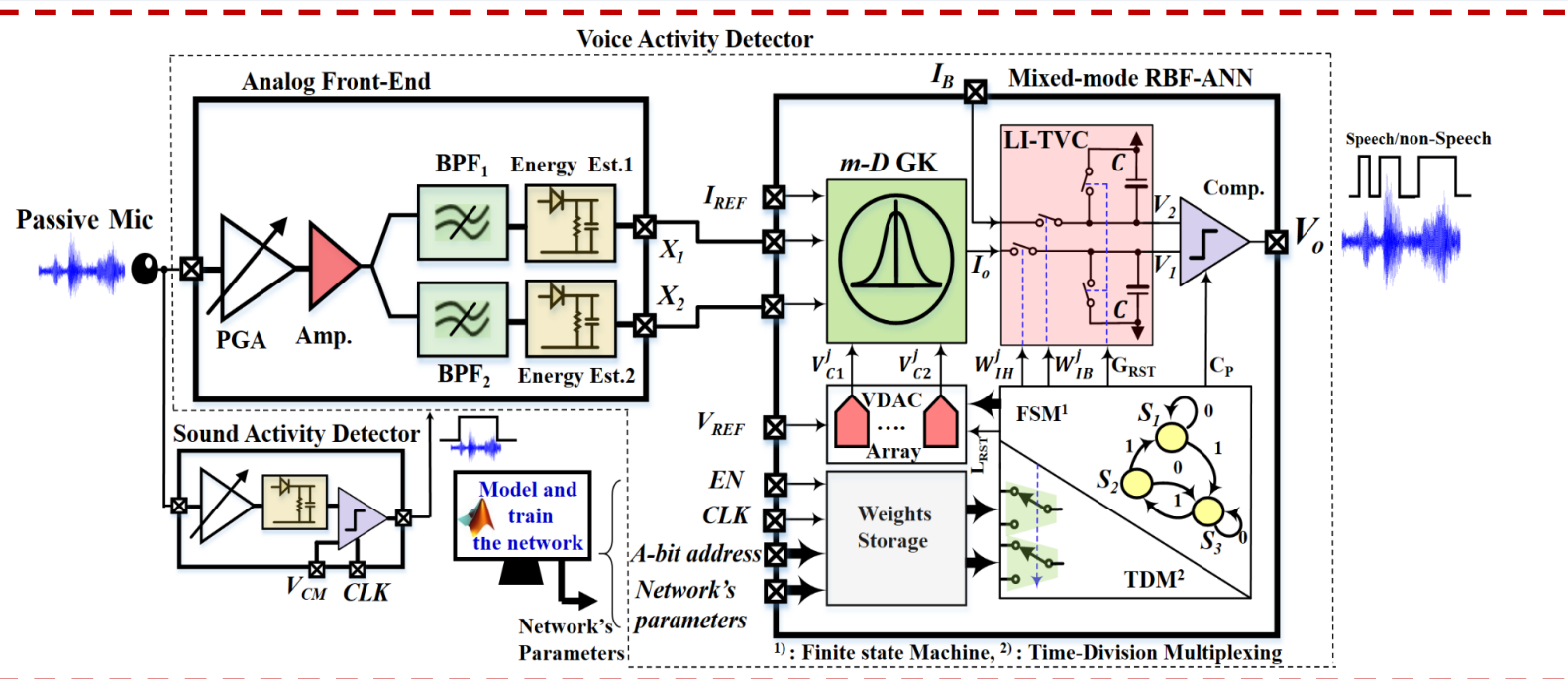
A. Reda



Zagazig Univ.
& MOSIS (Granted)

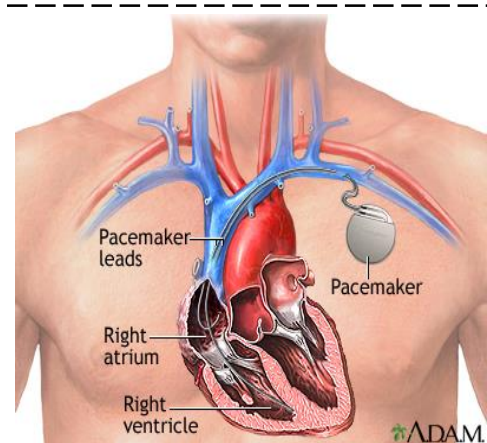
Item	Discrete Element	Very large scale Integrated Circuit
Area	Large	Small
Configuration	Easy	Complex
Price	Cheap	Expensive
Applications	Small production	Mass production
Power consumption	High	Low

Course Long-Term-Vision (2/3)

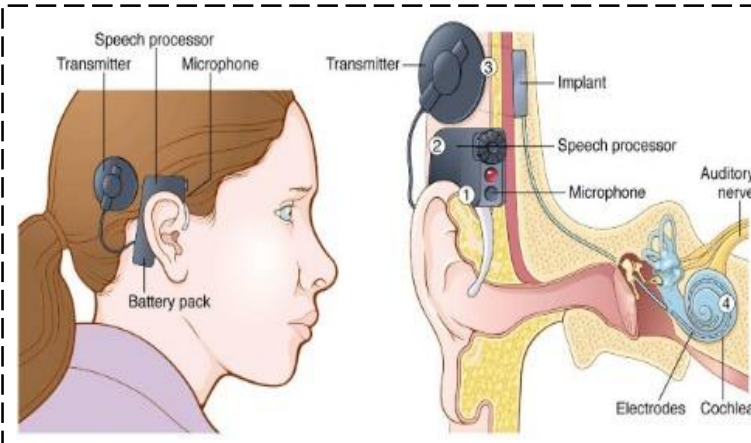




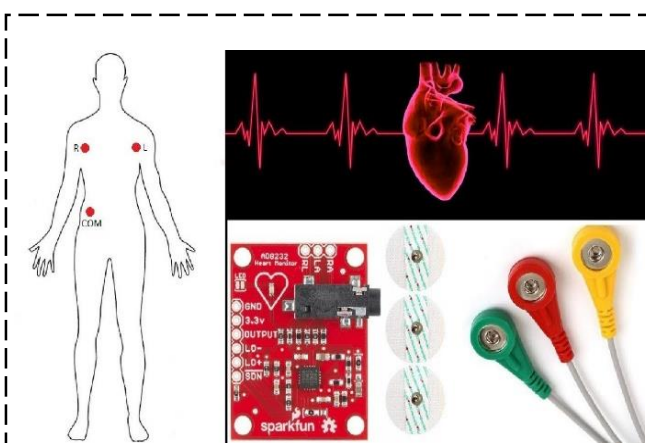
Electronics in Bio-medical Devices



Pacemaker



Cochlear implant

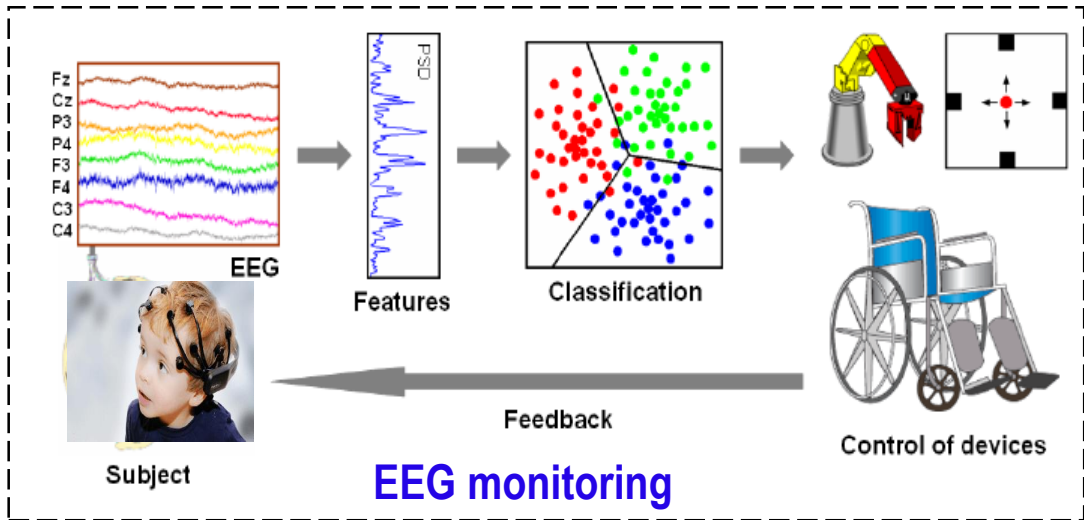


ECG monitoring

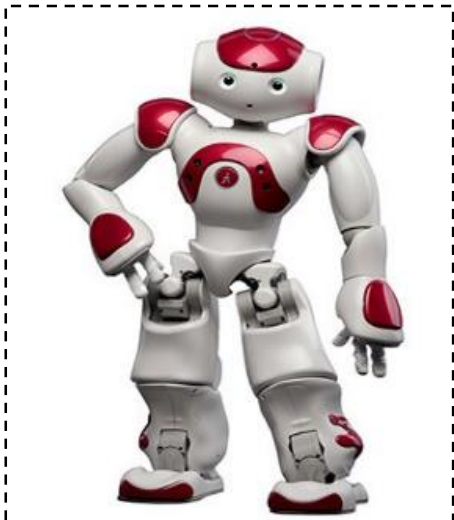
ECG	ElectroCardio-Gram
PPG	Photo-Plethy-smo Graphy
EEG	Electro-encephalo- gram



PPG monitoring



EEG monitoring





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Time_Plan and Project (1/2)



Happy new academic year
Curriculum introduction

1st week

Quiz1

2nd week

3rd week

Quiz2

4th week

Assignemnt1

5th week

Quiz3

6th week

7th week

Midterm

8th week

9th week

Quiz4

10th week

Project

11th week

Quiz5

12th week

13th week

Quiz6

14th week

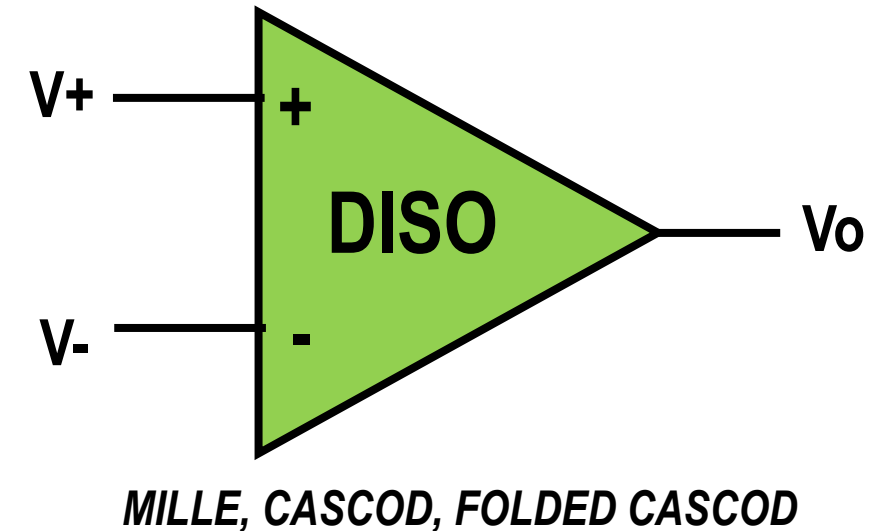
Final Exam

15th week



□ Design an trans-conductance amplifier.

Items	Specification
Open Loop Gain	> 60 dB
GBW (MHz)	> 1 MHz
Input CMR (Volts)	Maximum
Slew Rate (V/μsec)	>10 V/μ sec
P_{diss} (mW)	< 1 mW
Vout range (V)	Maximum
V_o off-set	Minimum
Phase margin (degrees)	> 45°
Output resistance	Maximum
CMRR, PSRR+, PSRR-	>70 dB



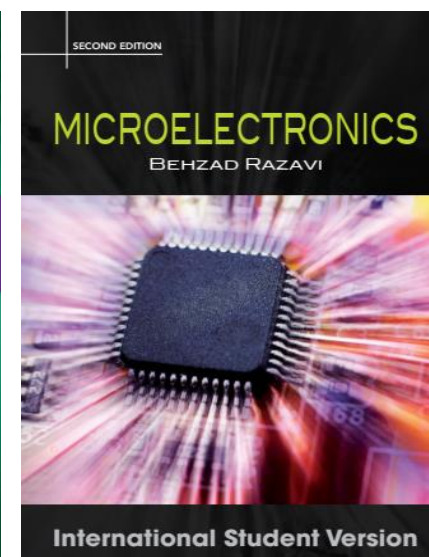
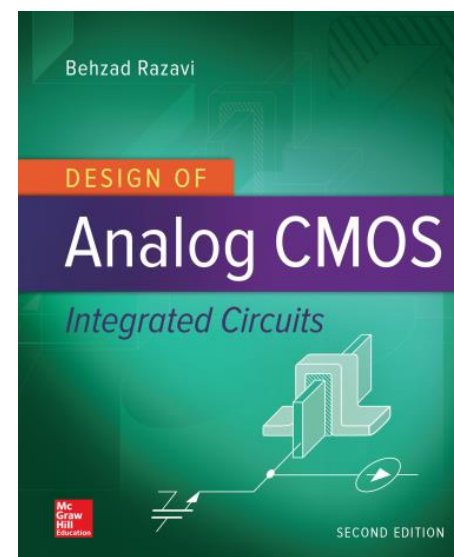
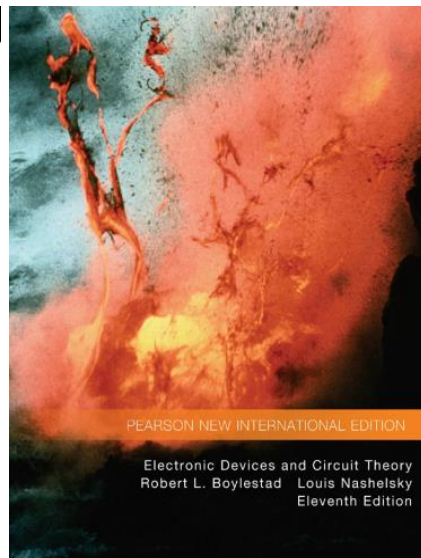
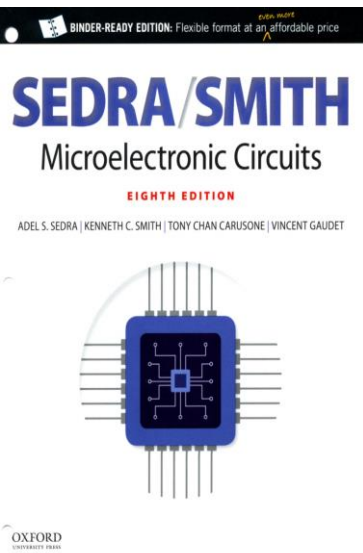


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References



- ☐ [1] A. S. Sedra et. al., **Microelectronic Circuits**, 8th edition, Oxford Series in Electrical and Computer Engineering, 2020
- ☐ [2] R. L. Boylestad, and L. Nashelsky, **Electronic Devices and Circuit Theory**, 11th © Pearson Education Limited 2014.
- ☐ [3] B. Razavi , **Microelectronics** , 2nd edition, John Wiley & Son, 2015 .
- ☐ [4] B. Razavi, **Design of Analog CMOS Integrated Circuits**, 2nd edition, McGraw-Hill Education, 2017
- ☐ [5] **Lectures Notes**



 **Stay on target**
[100%]

- ☐ **Lecture**
- ☐ **Quiz**
- ☐ **Assignment**
- ☐ **Project**

Thanks!